# Dependency-Preserving Evidence Packing for Numerical Question Answering

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Big Data course project ▪ FinQA, ConvFinQA, TAT-QA, HybridQA, TAT-HQA, MultiHiertt

# A number means nothing without its labels

Speaker: A

| company | payments vol. (bn) | transactions (bn) |
|---|---|---|
| visa inc. | $ 2457 | 50.3 |
| mastercard | 1697 | 27.0 |
| american express | 637 | 5.0 |
| discover | 102 | 1.6 |

Real FinQA development example. Gold program: `divide(637, const_5)` = 127.4

*"What is the average payment volume per transaction for American Express?"*

- The answer needs **two cells** plus the **row label** and **two column headers**.
- Given "637" and "5.0" alone, the model cannot tell which is which.
- With a small reading budget, labels compete with facts for space.

# Research question and data

Speaker: A

## Question

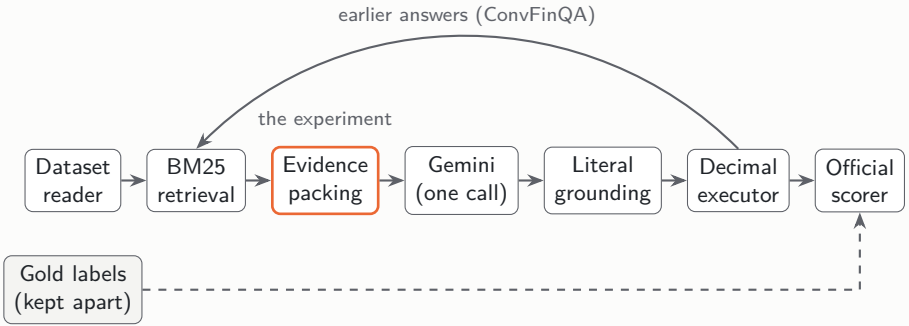
With the **same model** and the **same evidence budget**, does keeping each fact's headers, units, hierarchy and links with it improve answers over ordinary packing?

| | Dataset | Dev questions |
|---|---|---|
| Assigned | FinQA | 883 |
| | ConvFinQA | 1,490 turns |
| | TAT-QA | 1,668 |
| | HybridQA | 3,466 |
| Added | TAT-HQA | 824 |
| | MultiHiertt | 1,044 |

- One shared pipeline; only readers and scorers differ per dataset.
- Official evaluators, run unchanged at pinned versions.
- Related datasets share documents, so results are **never pooled**.
- HybridQA is general knowledge: a transfer test, not finance.

# One pipeline for six datasets

Speaker: A




- Each fact stores its text and the IDs of the facts needed to read it.
- Labels never reach the prompt: 0 of 443 inspected prompts contained one.
- Every raw response is saved, so runs resume and re-score without new calls.
- Spark builds the deduplicated fact store, checked against Python.

# Three ways to pack the same budget

Speaker: A

## Ordinary (O)

```
amex | payments | 637
amex | txns | 5.0
```

Labels repeated inline per cell. A fair baseline, not a weakened one.

## Dependency (D)

```
company | payments | txns
amex | 637 | 5.0
```

Recursive closure of headers, units, hierarchy and links, stored once.

## Length-matched (L)

```
637  5.0  102  1.6
```

Neighbouring facts up to D's length. Tests whether *more text* is enough.

### Literal grounding: the model copies values, code finds them

The model writes `divide("637", "5.0")`. We locate each value in the packed evidence or the question; a value found nowhere rejects the answer. The earlier design asked for character offsets and reached **2 of 883** FinQA questions. That run also used a different model, so we do not credit the format alone.

# A fair experiment on a free-tier budget

Speaker: B

## Budget must actually bind

- At 4,000 tokens, four datasets were not truncated in a 40-question sample: methods would differ only in layout.
- Caps chosen from document sizes, without labels: 400–1,500 tokens.

## Three measurements

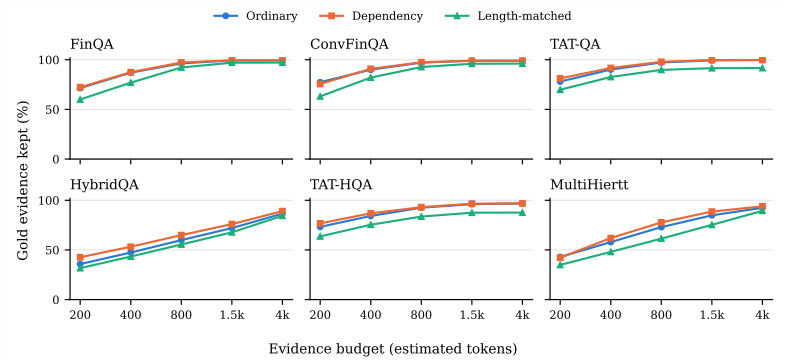
- **Coverage**: does gold evidence survive packing? All 9,268 questions, no model.
- **Block A**: O vs D vs L answers, `gemini-3.5-flash-lite`.
- **Block B**: executor vs direct answering, `gemini-3.1-flash-lite`.

## Rules

- One model per comparison; models are never compared with each other.
- Samples fixed by seed before any call; conversations kept whole.
- Failures stay in the denominator as zero.
- Paired bootstrap over whole documents, 2,000 resamples.
- 500 requests per day per model: runs stop and resume from saved answers.

# Result 1: dependency packing keeps more gold evidence

Speaker: B




Exact counts over every scorable development question. Questions kept only by D vs only by O at the chosen cap: HybridQA **146 vs 7**, MultiHiertt **42 vs 3**, TAT-QA 41 vs 15, TAT-HQA 33 vs 12, FinQA 9 vs 1, ConvFinQA 6 vs 1. At 200 tokens the closure overhead costs coverage on ConvFinQA and MultiHiertt.

# Result 2: effect on answers

Speaker: B

## Block A: packing (%)

| Dataset | $n$ | O | D | L | D−O | 95% CI |
|---|---|---|---|---|---|---|
| FinQA | 50 | 62.0 | 64.0 | 66.0 | +2.0 | [+0.0, +6.2] |
| ConvFinQA | 45 | 73.3 | 73.3 | 68.9 | +0.0 | [−6.0, +6.4] |
| TAT-QA | 50 | 60.2 | 71.5 | 69.4 | +11.3 | [+4.1, +19.9] |
| HybridQA | 50 | 54.9 | 60.1 | 49.8 | +5.3 | [−2.4, +13.0] |
| TAT-HQA | 50 | 56.7 | 60.7 | 52.7 | +4.0 | [−6.4, +14.5] |
| MultiHiertt | 50 | 47.0 | 56.8 | 41.0 | +9.8 | [+0.0, +20.0] |

## Block B: executor vs direct (%)

| Dataset | $n$ | Executor | Direct | Diff. | 95% CI |
|---|---|---|---|---|---|
| FinQA | 40 | 62.5 | 67.5 | −5.0 | [−17.5, +7.0] |
| ConvFinQA | 36 | 75.0 | 75.0 | +0.0 | [−8.6, +7.3] |
| TAT-QA | 40 | 65.2 | 58.7 | +6.5 | [−7.6, +20.7] |
| HybridQA | 40 | 50.2 | 66.6 | −16.4 | [−30.6, −2.0] |
| TAT-HQA | 40 | 60.0 | 56.7 | +3.3 | [−12.0, +19.3] |
| MultiHiertt | 40 | 63.7 | 41.7 | +22.1 | [+2.5, +40.0] |

- Packing: D $\geq$ O on all six datasets; the interval excludes zero only on **TAT-QA (+11.3)**.
- Intervals are wide at these sample sizes; six comparisons per block are uncorrected.
- Only robust Block B gap: **direct answering wins on HybridQA**, which has short text answers and no arithmetic.
- MultiHiertt's Block B row is official F1; under one scoring rule for both sides the gap is +6.2 (next slide).

# What failed, and what we learned

Speaker: B

## A result we withdrew

- MultiHiertt first showed the executor **+22.1** points ahead.
- Direct answers wrote −29.92 for a gold −0.2992; the official span path marks that wrong.
- Same scoring rule for both sides: **+6.2**, interval [−6.2, +20.0]. No claim.

## Where answers still go wrong

- Numbers: wrong values, reversed percentage-change direction.
- Text: over-long or wrong spans (HybridQA, TAT-QA).
- Validation failures are now rare.

**Uncredited executor answers, Block B**

| Dataset | Val | Sign | Span | Grnd | Abst | Other | All |
|---|---|---|---|---|---|---|---|
| FinQA | 7 | 3 | 0 | 2 | 4 | 1 | 17 |
| ConvFinQA | 6 | 2 | 0 | 1 | 0 | 0 | 9 |
| TAT-QA | 2 | 2 | 13 | 0 | 3 | 0 | 20 |
| HybridQA | 0 | 0 | 19 | 2 | 4 | 0 | 25 |
| TAT-HQA | 9 | 1 | 2 | 3 | 1 | 0 | 16 |
| MultiHiertt | 6 | 2 | 2 | 0 | 3 | 2 | 15 |

# Big Data, and conclusions

Speaker: B

## Spark preparation

- Partitioned Parquet store: 1.36 M facts, 1.88 M dependency edges, 0 unresolved.
- Finds 491 identical-content document pairs across datasets.
- Every run must equal the Python reference.
- Honest timing: Python takes 15.8 s; Spark 37.0 s at best (4 cores). At 1.4 M rows, Spark's value is the checked store, not speed.

## Conclusions

- Keeping labels with facts preserves more needed evidence, most when documents far exceed the budget.
- Answers move the same way on all six datasets, but only TAT-QA's gain is clear at $n = 50$.
- Copy-and-resolve answers replaced a pipeline that almost always failed validation (the model also changed).
- An executor gives auditability, not a guaranteed accuracy gain.

Limitations: small samples, inferred headers, estimated token counts, hosted models that can change.